Geo-Text-Vision Alignment for Perovskite Property Prediction

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Abstract

We present Geo-Text-Vision (GTV), a three-modality framework for band gap prediction of perovskite oxides (ABO_3) that jointly exploits crystal structure, natural-language descriptions, and rendered images. Each material is represented by a crystal graph processed by CGCNN, a Robocrystallographer-generated text description encoded by SciBERT, and a ball-and-stick image encoded by a frozen ResNet18. We introduce an asymmetric FiLM fusion in which text semantically modulates the structural representation while image features are concatenated at the fusion head, paired with a multi-task objective that adds metallicity classification as an auxiliary regularizer in the small-data regime. On a curated dataset of 592 ABO_3 perovskites from the Materials Project, our best model achieves a 5-seed ensemble test MAE of 0.4862 eV—an 18% improvement over the structure-only CGCNN baseline and 26% over the text-only SciBERT baseline. We systematically compare fusion strategies (concatenation, gated, FiLM, cross-attention), warm vs. cold start initialization, and an ALBEF-style contrastive pretraining stage that reduces per-seed variance without sacrificing mean accuracy. A random-embedding ablation disentangles semantic content from architectural capacity, showing that each contributes roughly half of the total gain over the unimodal baseline. Finally, we demonstrate a zero-shot LLM-in-the-loop inverse design agent that reuses the trained ensemble as an oracle to propose candidate materials for user-specified target band gaps without any retraining, closing the forward–inverse loop in under a minute.

1 Introduction

Perovskite oxides (ABO_3)—comprising an A-site metal, a B-site transition metal, and three oxygens—underpin a wide range of energy-critical technologies, including next-generation solar cells, LEDs, and photocatalysts [Xie and Grossman, 2018]. Their tunability arises from the chemical flexibility of the ABO_3 skeleton: subtle cation substitutions or octahedral distortions can induce dramatic shifts in electronic behavior, spanning from metallic to wide-gap insulating regimes. This versatility makes perovskites attractive for materials design, but also creates a vast combinatorial space that is impractical to explore via trial-and-error experiments or computationally expensive DFT, which requires several CPU hours per material evaluation.

Machine learning has emerged as the natural accelerator for materials discovery. GNNs such as CGCNN [Xie and Grossman, 2018] and MEGNet [Chen et al., 2019] encode crystal structures as graphs and achieve strong predictive performance, but are largely black-box and struggle to capture high-level chemical semantics. Language models such as SciBERT [Beltagy et al., 2019] and MatSciBERT [Gupta et al., 2022] extract chemically meaningful patterns from scientific text, offering interpretability but relying on human-written corpora and lacking explicit geometric grounding for novel or hypothetical materials. This separation highlights a key gap: while multimodal fusion is mature in vision-language tasks, it has not been systematically applied to materials property

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prediction. Neither structural nor textual representations alone capture the full picture, and their complementary nature suggests that fusion should outperform either modality.

In this paper, we propose Geo-Text-Vision (GTV), a three-modality framework for perovskite band gap prediction. We represent each crystal from three complementary views: (1) a crystal graph processed by CGCNN, capturing precise local geometry; (2) an auto-generated natural-language description processed by SciBERT, encoding semantic concepts such as coordination environments and octahedral tilting motifs produced by Robocrystallographer [Ganose and Jain, 2019]; and (3) a rendered ball-and-stick image processed by a frozen ResNet18 [He et al., 2015], capturing global crystal symmetry and topology. We fuse these modalities through asymmetric FiLM [Perez et al., 2018] in which text conditions the structure graph representation, while image features enter at the fusion head to bound the overfitting risk from a frozen visual encoder in the small-data regime. Multi-task learning with an is-metal auxiliary head provides free regularization. We further demonstrate an LLM-in-the-loop inverse design agent that reuses the trained forward model as an oracle to iteratively search for perovskites with user-specified target band gaps.

Our main contributions are: (1) a trimodal fusion architecture (GTV) with asymmetric FiLM modulation and multi-task learning that achieves 18% MAE improvement over the structure-only baseline; (2) a systematic study of fusion strategies (concatenation, gated, FiLM, cross-attention), initialization strategies (warm vs. cold start, ALBEF), and a modality ablation that disentangles semantic from architectural gains; and (3) a zero-shot LLM-in-the-loop inverse design agent that closes the forward-inverse loop without retraining.

2 Related Work

Graph-based property prediction. GNNs have become the dominant approach for structure-based materials property prediction. CGCNN [Xie and Grossman, 2018] demonstrated accurate structure–property learning from atomic coordinates. MEGNet [Chen et al., 2019] incorporated global state variables. ALIGNN [Choudhary and DeCost, 2021] improved accuracy by including bond angles via line-graphs. More recently, JMP [Shoghi et al., 2024] achieved strong generalization through large-scale pretraining across molecular and crystalline datasets. Despite high accuracy, these models are largely black boxes and struggle to capture high-level chemical semantics such as coordination environments or octahedral distortion motifs.

Text-based modeling. NLP models extract chemical knowledge from literature or SMILES strings. MatSciBERT [Gupta et al., 2022] and MatBERT [Trewartha et al., 2022] are domain-specific BERT models pretrained on materials science text. T5Chem [Lu and Zhang, 2022] and GPTChem [Jablonka et al., 2024] operate on SMILES for prediction and generation tasks. LLM-Prop [Rubungo et al., 2023] showed that LLMs can predict crystal properties from textual structural descriptions generated by Robocrystallographer [Ganose and Jain, 2019]. However, these models lack explicit geometric grounding and are limited for novel or hypothetical materials.

Multimodal representation learning. Computational materials research has adapted the contrastive vision-language alignment method pioneered by CLIP [Radford et al., 2021]. A notable recent paper introduced MultiMat [Moro et al., 2025], which uses contrastive learning to align crystal structures, electronic properties, and text, improving downstream property prediction. Existing approaches emphasize global alignment rather than fine-grained multimodal fusion on individual compounds for specific material families, and none combines all three modalities (graph, text, image) in a unified regression framework. The efficacy of align-before-fuse paradigm [Li et al., 2021] has not yet been demonstrated within the domain either.

Image modality for crystals. Rendered crystal images contain information about global symmetry and topology complementary to local graph convolutions. ResNet-based encoders pretrained on ImageNet [Russakovsky et al., 2015] have been repurposed for materials applications, but only in a unimodal manner [Zhang et al., 2023, Tiong et al., 2020, Ra et al., 2021]. In our setting, a frozen ImageNet prior extracts useful global shape information without fine-tuning, provided the new projection head is zero-initialized to avoid regressing the unimodal baseline.

LLMs for inverse materials design. Gruver et al. [2024] generated inorganic crystal structures from string encodings using fine-tuned LLMs. Takahara et al. [2025] combined LLM reasoning with generative models in iterative design loops. These frameworks focus on generation and decisionmaking, but has limited exploration of efficiet uses of multimodal material representations. We complement this line of work by reusing a trained multimodal property prediction model as an oracle within a zero-shot LLM-driven search using Qwen2.5-VL [Wu et al., 2025], targeting property-conditioned material design without any retraining.

Our contribution. To our knowledge, this work is the first to systematically compare fusion strategies for three-modality (graph, text, image) material property prediction using *machine-generated* text descriptions and rendered images, and to disentangle semantic from architectural contributions through random embedding ablation. We focus on a low-diversity ABO_3 perovskite setting where subtle structural variations challenge graph-only models, providing a controlled testbed for multimodal evaluation. Beyond prediction, we bridge the framework toward inverse design via an agentic LLM loop.

3 Problem Statement

Let $\mathcal{D} = \{(G_i, T_i, V_i, y_i, m_i)\}_{i=1}^{N}$ denote a dataset of N perovskite materials, where: G_i is the crystal structure as a graph (CIF-derived), with nodes as atoms and edges encoding interatomic interactions; T_i is a natural-language description generated by Robocrystallographer; V_i is a rendered ball-and-stick image (224×224 dual-view PNG); $y_i \in \mathbb{R}_{\geq 0}$ is the band gap in eV; and $m_i \in \{0, 1\}$ is the metallicity label ($m_i = 1$ iff band gap $= 0$).

Our primary objective is to learn $f_\theta : (G, T, V) \mapsto \hat{y}$ that minimizes the Huber regression loss ($\delta = 0.5$):

$$\mathcal{L}_{\text{reg}}(\theta) = \frac{1}{N} \sum_{i=1}^{N} \ell_\delta\big(f_\theta(G_i, T_i, V_i),\, y_i\big).$$

We jointly optimize a multi-task objective with an auxiliary binary metallicity head:

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{reg}}(\theta) + \lambda\, \mathcal{L}_{\text{metal}}(\theta), \quad \lambda = 0.3.$$

Our hypothesis is that all three modalities encode complementary information:

$$\mathcal{L}(f_\theta(G, T, V)) \;<\; \min\big\{\mathcal{L}(f_G(G)),\, \mathcal{L}(f_T(T)),\, \mathcal{L}(f_V(V))\big\}.$$

4 Proposed Method

Figure 1 illustrates the overall architecture. We pair three unimodal encoders with an asymmetric FiLM fusion module and two task heads.

4.1 Unimodal Encoders

4.1.1 Structure Encoder: CGCNN

We adopt CGCNN [Xie and Grossman, 2018] as the structure encoder f_G. Each atom i is represented by an 82-dimensional one-hot feature vector $\mathbf{v}_i$, linearly projected to $\mathbb{R}^{d_a}$. Interatomic bonds within 8 Å are encoded via $K = 41$ Gaussian basis functions (centers 0–8 Å, step 0.2 Å). Each atom aggregates messages from its $M = 12$ nearest neighbors through $L = 3$ graph convolution layers:

$$\mathbf{h}_i^{(\ell+1)} = \mathbf{h}_i^{(\ell)} + \text{softplus}\left(\text{BN}\left(\sum_{j \in \mathcal{N}(i)} \sigma(\mathbf{z}_{ij}^{\text{filter}}) \odot \text{softplus}(\mathbf{z}_{ij}^{\text{core}})\right)\right),$$

where $[\mathbf{z}_{ij}^{\text{filter}}; \mathbf{z}_{ij}^{\text{core}}] = \text{BN}(\mathbf{W}^{(\ell)}[\mathbf{h}_i^{(\ell)}; \mathbf{h}_j^{(\ell)}; \mathbf{e}_{ij}] + \mathbf{b}^{(\ell)})$ and σ is the sigmoid function. The crystal-level embedding $h_G \in \mathbb{R}^{128}$ is obtained by mean pooling over atoms followed by a fully connected layer with softplus activation.

4.1.2 Text Encoder: SciBERT

We use SciBERT [Beltagy et al., 2019], pretrained on 1.14M scientific papers, as the text encoder f_T. Given a Robocrystallographer description T, we extract the `[CLS]` token embedding $h_T \in \mathbb{R}^{768}$. During fusion training, the last two encoder layers and the pooler are unfrozen for end-to-end fine-tuning.

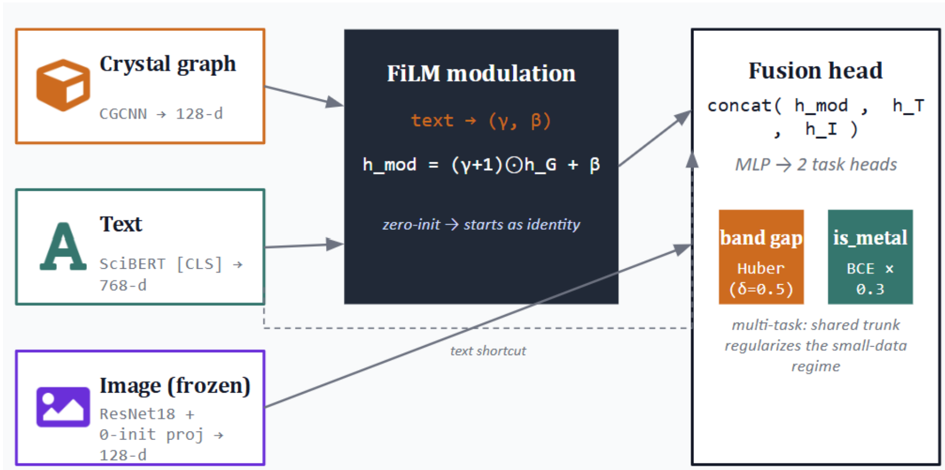

Figure 1: GTV architecture. Text (SciBERT) generates FiLM modulation parameters $(\boldsymbol{\gamma}, \boldsymbol{\beta})$ that condition the structure graph representation (CGCNN). The image branch (frozen ResNet18 + zero-init MLP) concatenates into the fusion head only. Two task heads—band gap regression (Huber) and metallicity classification (BCE $\times 0.3$)—branch from a shared MLP.

4.1.3 Image Encoder: Frozen ResNet18

We render each crystal structure as a 224×224 dual-view ball-and-stick PNG using ASE. A ResNet18 [He et al., 2016] pretrained on ImageNet encodes the image with *frozen* parameters throughout training, and a two-layer MLP projects the 512-d pooled features to $h_V \in \mathbb{R}^{128}$ with *zero-initialized* weights. Zero-initialization ensures the image branch contributes zero at step 0; the optimizer recruits it only if the gradient signal favors it, eliminating regression risk on the unimodal baseline.

4.2 Fusion Methods

4.2.1 Asymmetric FiLM Fusion

We use Feature-wise Linear Modulation (FiLM) [Perez et al., 2018] to reflect the *asymmetric* roles of the modalities. Text provides high-level semantic context ("corner-sharing octahedra," "distorted") that modulates the geometric precision encoded by the GNN; the GNN should not reciprocally modulate the text, as local geometry is the ground truth and text is a semantic gloss. The FiLM generator maps $h_T \in \mathbb{R}^{768}$ to modulation parameters via a two-layer MLP:

$$(\boldsymbol{\gamma}, \boldsymbol{\beta}) = \mathrm{split}(\mathbf{W}_2 \cdot \mathrm{GELU}(\mathbf{W}_1 h_T + \mathbf{b}_1) + \mathbf{b}_2), \quad \boldsymbol{\gamma}, \boldsymbol{\beta} \in \mathbb{R}^d,$$

where $\mathbf{W}_2$ is zero-initialized so modulation starts as identity ($\boldsymbol{\gamma} = \mathbf{1}, \boldsymbol{\beta} = \mathbf{0}$). The modulated structure embedding is:

$$h_{\text{mod}} = \mathrm{LN}((\boldsymbol{\gamma} + \mathbf{1}) \odot h'_G + \boldsymbol{\beta}) + h'_G,$$

where $h'_G = \mathrm{GELU}(\mathrm{LN}(\mathbf{W}_G h_G)) \in \mathbb{R}^d$ and the residual skip ensures gradient flow when $\boldsymbol{\gamma} \approx \mathbf{0}$.

The image embedding is excluded from FiLM modulation—coupling a frozen visual prior (pretrained on natural images) to the structure features would introduce uncontrolled variance with no quality guarantee. Instead, h_V is concatenated at the fusion head:

$$h_{\text{fused}} = \mathrm{concat}(h_{\text{mod}},\ h'_T,\ h_V) \in \mathbb{R}^{d+d+128},$$

where $h'_T = \mathrm{GELU}(\mathrm{LN}(\mathbf{W}'_T h_T))$ is a text shortcut. Two task heads branch from an MLP applied to h_{fused}: a band gap regression head (Huber loss, $\delta = 0.5$) and a metallicity classification head (BCE).

4.2.2 Cross-Attention Fusion

As an alternative to FiLM's global-vector modulation, we implement token-level cross-attention fusion that enables fine-grained atom-to-token interactions. Rather than pooling the CGCNN output to a single crystal vector before fusion, we expose the per-atom embeddings $\{\mathbf{h}_i^{(L)}\}_{i=1}^{N_a}$ from the last graph convolution layer as queries, and use the full sequence of SciBERT token embeddings $\{\mathbf{t}_j\}_{j=1}^{N_t} \in \mathbb{R}^{768}$ as keys and values.

Each atom embedding is first projected to the attention dimension: $\mathbf{q}_i = \mathbf{W}_Q \mathbf{h}_i^{(L)} \in \mathbb{R}^{d_k}$, and the token embeddings are projected to keys and values: $\mathbf{K} = \mathbf{T}\mathbf{W}_K \in \mathbb{R}^{N_t \times d_k}$, $\mathbf{V} = \mathbf{T}\mathbf{W}_V \in \mathbb{R}^{N_t \times d_v}$. The cross-attention output for atom i is:

$$\tilde{\mathbf{h}}_i = \text{softmax}\left(\frac{\mathbf{q}_i \mathbf{K}^\top}{\sqrt{d_k}}\right) \mathbf{V} \ \in \mathbb{R}^{d_v}.$$

The attended atom features are mean-pooled over atoms within each crystal to obtain $h_{\text{cross}} = \frac{1}{N_a} \sum_i \tilde{\mathbf{h}}_i \in \mathbb{R}^{d_v}$, which is then concatenated with the text `[CLS]` embedding h_T and passed through the prediction MLP. Multi-head attention with $H = 4$ heads is used throughout.

4.2.3 Multi-task Learning as Implicit Regularization

The metallicity auxiliary head forces the shared trunk to learn features that distinguish metals from non-metals—a free supervisory signal that regularizes band gap regression on our 383-sample training set. Both the metal head weights and the FiLM $\boldsymbol{\gamma}, \boldsymbol{\beta}$ generator are zero-initialized, so the multi-task objective starts from the same landscape as the single-task loss, preserving stable early-training dynamics.

4.2.4 Small-data Engineering Strategies

Four engineering choices beyond the architecture proved critical.

Cold start, not warm start. Initializing multi-task FiLM from a pre-existing single-task FiLM checkpoint ("warm start") degraded performance by 9.5% relative to random initialization (Exp 5 warm: 0.5388 vs. Exp 5b cold: 0.4906 eV). A checkpoint optimal under loss $\mathcal{L}_1$ is a saddle under $\mathcal{L}_1 + \lambda\mathcal{L}_2$; warm-starting the wrong objective is worse than cold start with shared-encoder pretraining.

5-seed ensembling. On a 56-sample test set, the standard deviation of single-seed MAE is $\sim$0.05 eV, comparable to the performance gaps between models. We ensemble 5 independently trained seeds by averaging band gap predictions. A "SoTA" claim based on a single seed (original Exp 3 FiLM: 0.5094 eV) was revised to 0.5379$\pm$0.024 eV on a 5-seed re-run—a 5.6% discrepancy from a lucky draw.

Contrastive align-before-fuse (ALBEF) initialization. Inspired by CLIP [Radford et al., 2021] and ALBEF [Li et al., 2021], we pretrain a contrastive alignment stage using symmetric InfoNCE:

$$\mathcal{L}_{\text{InfoNCE}} = \frac{1}{2}\big[\text{CE}(\tau \cdot z_G z_T^\top, \mathbf{I}) + \text{CE}(\tau \cdot z_T z_G^\top, \mathbf{I})\big],$$

where τ is a learnable temperature and z_G, z_T are L2-normalized 128-d projections. Using the pretrained encoder states as initialization for the multi-task FiLM does not improve the mean MAE, but significantly reduces per-seed standard deviation (σ: 0.056 $\rightarrow$ 0.038 from Exp 5b to Exp 7), yielding a more reliable deployment model.

4.3 Inverse Design Agent

We build an LLM-in-the-loop inverse design agent on top of the 5-seed MT-FiLM ensemble (Exp 5b, test MAE 0.49 eV). The loop contains three components:

Oracle: The 5-seed ensemble evaluates candidate materials, returning band gap mean $\pm$ std and is-metal probability—no retraining required.

Proposer: Qwen2.5-VL-3B-Instruct [Wu et al., 2025] in text mode, zero-shot. Given a target band gap and full history of prior candidates with oracle predictions, the LLM proposes (A, B) element substitutions in strict JSON. Two consecutive parse failures trigger a uniform-random vocabulary fallback.

Loop: Iteration 0 samples random candidates; iterations 1–4 are LLM-conditioned on the full history. We run 5 candidates per iteration across 3 target band gaps (1.0, 2.0, 3.0 eV), for 75 oracle evaluations total. Atom substitution applies `pymatgen.Structure.replace_species` to the base CIF and regex-rewrites the Robocrystallographer text in-place. A charge-balance heuristic ($n_A + m_B = 6$) flags chemically implausible proposals.

5 Experimental Methodology

5.1 Dataset and Data Splits

We curate 592 ABO_3 perovskite oxides from the Materials Project [Jain et al., 2013] API, filtered by strict stoichiometry ($A_1B_1O_3$) and restricted to six common perovskite space groups (Pnma, Pm$\bar{3}$m, R$\bar{3}$c, I4/mcm, P4mm, R3c). The dataset spans 62 A-site and 70 B-site elements. Each material has three modalities: (1) a crystal structure in CIF format; (2) a natural-language description generated by Robocrystallographer [Ganose and Jain, 2019] in `fast_no_mineral` mode, successfully generated for 493/592 structures (83.3%); and (3) a 224×224 dual-view ball-and-stick PNG rendered via ASE.

The prediction target is the band gap (eV), ranging from 0 to 6.09 eV (mean 1.61 ± 1.71 eV), with 216 metals and 376 non-metals. We split data using a deterministic hash-based 80/10/10 partition (SHA-256 on `material_id` with a fixed seed), yielding **383/54/56** train/val/test for all text-dependent experiments. All models are evaluated using mean absolute error (MAE, eV, ↓), RMSE (eV, ↓), and R^2 (↑); all reported numbers are 5-seed ensemble metrics.

5.2 Training Details

All experiments run on a single NVIDIA V100 or L40S GPU. For CGCNN (Exp 1): Adam, lr= 10^{-3}, wd= 10^{-4}, batch 64, ReduceLROnPlateau (patience 15, factor 0.5). For SciBERT (Exp 2): AdamW, lr= 2×10^{-5}, wd= 0.01, batch 16. For all fusion experiments (Exp 3–8): AdamW, lr= 10^{-4}, wd= 0.01, batch 16, cosine LR with 10-epoch warmup, gradient clipping (max norm 1.0), AMP fp16, early stopping on validation MAE (patience 20–30 epochs). For contrastive alignment (Exp 4/7): batch 128, lr 5×10^{-4} for projection heads, $0.1\times$ for encoders. Each fusion experiment runs 5 independent seeds with ensemble averaging at inference.

6 Results and Discussion

6.1 Main Property Prediction Results

Table 1 presents 5-seed ensemble results for all eight experiments. The structure-only CGCNN baseline achieves ensemble MAE 0.5921 eV ($R^2 = 0.682$) and text-only SciBERT achieves 0.6566 eV ($R^2 = 0.624$), confirming that graph representations have an edge over text alone for geometric property prediction. All fusion configurations outperform both unimodal baselines.

These results also suggest that modality interaction matters more than raw representational power at this dataset scale. The text branch alone underperforms the structural baseline, yet conditioning the graph encoder on textual information consistently improves prediction. This indicates that the textual modality is not acting as an independent predictor, but rather as a contextual prior that reshapes how structural features are interpreted. In practice, descriptions extracted from CIF files encode chemically meaningful concepts such as bonding environment, coordination geometry, and symmetry that are difficult for shallow graph message passing to recover explicitly from connectivity alone.

6.2 Fusion Strategy Analysis

FiLM vs. cross-attention. Among bimodal fusion strategies, FiLM (0.5173 eV) outperforms cross-attention (0.5395 eV) on the test set. Cross-attention adds token-level atom-to-token interactions and ∼10% more parameters, but additional capacity leads to overfitting at this scale. FiLM's asymmetric design uses text to modulate the primary modality of structure, rather than treating both as interchangeable roles. This design is well-suited for the nature of structure-property modeling, and provides a cleaner inductive bias that generalizes well towards unseen data points.

Warm start vs. cold start. Initializing multi-task FiLM from the single-task FiLM checkpoint degrades ensemble MAE by 9.5% relative to cold start (0.5388 vs. 0.4906 eV). A local optimum

Table 1: Band gap prediction results (5-seed ensemble on the test set). Δ is relative MAE improvement over the CGCNN baseline. Best in **bold**.

Model	**Modality**	**Ens. MAE (eV)** $\downarrow$	**Seed** σ	Δ
SciBERT (finetune)	Text only	0.6566	0.022	-10.9%
CGCNN	Structure only	0.5921	0.023	0%
FiLM	Struct+Text	0.5173	0.024	+12.6%
MT-FiLM warm start	Struct+Text	0.5388	0.016	+9.0%
Cross-Attention	Struct+Text	0.5395	0.034	+8.9%
ALBEF init	Struct+Text	0.4990	0.038	+15.7%
MT-FiLM cold start	Struct+Text	0.4906	0.056	+17.1%
GTV 3-modality	**Struct+Text+Image**	**0.4862**	0.042	**+17.9%**

Table 2: Trimodal ablation experiments (5-seed ensemble test MAE). "rand" replaces the corresponding modality with random $\mathcal{N}(0, 1)$ noise of the same dimension.

Configuration	**Test MAE (eV)** $\downarrow$	**vs. Full Model**
Full model (struct + text + image)	**0.4862**	—
Text ablated (struct + rand + image)	0.5137	+5.7%
Image ablated (struct + text + rand)	0.5332	+9.7%
Both ablated (struct + rand + rand)	0.5394	+10.9%
CGCNN only (struct)	0.5921	+21.8%

under $\mathcal{L}_1$ is a saddle under $\mathcal{L}_1 + \lambda\mathcal{L}_2$. The cold-start model, initialized from independently pretrained CGCNN and SciBERT encoders, is strictly better. This indicates that initializing from a single-task FiLM checkpoint biases representation towards the original objective, making it harder for the model to reorganize its latent space when multitask losses are introduced.

Vision modality. Adding the image branch to the best bimodal model (0.4906 eV) yields the best overall result (0.4862 eV), a further 0.9% gain. The image encoder uses a frozen ImageNet prior with zero-initialized projection, contributing zero at step 0. The small but consistent gain confirms that rendered crystal images carry complementary information about global symmetry not captured by local graph convolutions or bond-level text descriptions.

An important observation is that all successful multimodal variants preserve the crystal graph as the dominant computational backbone. None of the experiments support the idea that text or images can replace atomistic structure representations for band gap prediction. Instead, the additional modalities function as weak-but-complementary sources of information that refine the structural latent space. This asymmetry is scientifically reasonable, as electronic properties ultimately arise from atomic arrangement and composition. The empirical results therefore favor structure-centered multimodality (FiLM) over more symmetric multimodal architectures (cross-attention).

6.3 Modality Ablation: Semantic Content vs. Architectural Capacity

A key concern in multimodal fusion is whether performance gains stem from semantically meaningful content or from additional model capacity and input dimensionality. To disentangle these, we replace text or image embeddings with random Gaussian noise $\mathcal{N}(0, 1)$ of the same dimensions as the actual embeddings, keeping the architecture fixed (Table 2).

The trimodal gain over unimodal CGCNN (-0.1059 eV) decomposes as follows:

- **Architectural capacity**: 0.5921 $\rightarrow$ 0.5394, **-0.0527 eV (MAE $\downarrow$ 8.9%)**, which is the gain from extra parameters and input dimensionality alone.
- **Semantic content**: 0.5394 $\rightarrow$ 0.4862 , **-0.0532 eV (MAE $\downarrow$ 10.9%)**, which is the gain from replacing random noise with real semantic embeddings.

Semantic information has greater contributions than architectural capacity alone in explaining the total gain. Notably, the image ablation (9.7% penalty) exceeds the text ablation (5.7%), indicating that the image branch carries more unique information in this configuration, despite using a far smaller

Table 3: Inverse design results. Best candidates (lowest oracle |error|) across 5 iterations × 5 candidates per target.

Target (eV)	Best Candidate	Pred ± std (eV)	|Error| (eV)	Charge OK?
1.0	$CeGeO_3$ (mp-19269)	0.787 ± 0.205	0.213	No
2.0	$NaPbO_3$ (mp-756464)	1.925 ± 0.325	**0.075**	No
3.0	$LaNbO_3$ (mp-1180739)	2.573 ± 0.537	0.427	Yes

encoder (frozen ResNet18 vs. 768-d SciBERT). This is consistent with the visual branch capturing global symmetry (space group, unit cell geometry) that local graph message-passing and bond-level text descriptions both underrepresent.

The ablation results also clarify the importance of decomposing multimodal improvements for genuine insights, as both semantic and architectural effects are genuinely present in the proposed model. For our task, the semantically meaningful content is still the leading factor in driving the overall improvements. However, simply increasing parameter count and feature dimensionality already yields notable observed gain even when semantic information is destroyed.

6.4 Contrastive Alignment and Training Stability

ALBEF initialization achieves ensemble MAE of 0.4990 eV (2%) above the cold-start MT-FiLM in expectation, but with substantially reduced per-seed standard deviation: σ drops from 0.056 to 0.038 (Table 1). Contrastive pretraining aligns the latent space in a way that makes downstream fine-tuning more stable across seeds, even though it does not directly optimize band gap prediction. This observation of "variance reduction without mean gain" is practically valuable, as a tighter prediction distribution is more reliable for deployment even when the average accuracy is comparable.

6.5 Inverse Design Results

Table 3 summarizes the best candidates found by the inverse design agent for three target band gaps. The agent closes the forward-inverse loop in 55 seconds (75 oracle evaluations) without any model retraining or fine-tuning.

Convergence behavior varies by target. For target 1.0 eV, the best-so-far oracle error decreases monotonically across iterations ($0.645 \rightarrow 0.213$ eV). For target 2.0 eV, the agent rapidly locks onto $NaPbO_3$ at iteration 1 (|error|=0.075 eV, approximately one-sixth of the ensemble's own test MAE), demonstrating that the LLM can exploit chemistry intuition effectively. For target 3.0 eV, performance is non-monotonic: $LaNbO_3$ (charge-balanced, |error|=0.427) is found at iteration 2, but the LLM subsequently oscillates to $BaTiO_3$, where the surrogate assigns $\sim$1.16 eV (vs. the LLM's chemistry prior of $\sim$3 eV). This disagreement between LLM knowledge and the surrogate oracle—neither side can win without DFT validation—illustrates a genuine limitation of surrogate-driven inverse design.

Vocabulary analysis shows the LLM concentrating B-site mass on Ti by iterations 3–4 for the 2.0 and 3.0 eV targets (d^0 Ti^{4+} is a known wide-gap B-site), while spreading across Ge, Pr, Sc, W, Zr for the 1.0 eV target. This zero-shot chemistry intuition without fine-tuning demonstrates that the LLM is conditioning meaningfully on the target property.

It's worth noting that two of three best candidates trigger charge-balance warnings. Therefore, further DFT-validation and experimental studies should be conducted before genuinely claiming the discovery of any new materials. Nonetheless, we demonstrated that the trained forward model is reusable as an oracle for property-targeted LLM-driven search without retraining, and such methodology would become quantitatively stronger and practically more promising with a higher-accuracy forward oracle and DFT-in-the-loop validation.

6.6 Discussion and Error Analysis

Prediction errors concentrate on materials with rare-earth B-site elements (Er, Eu, Ho, Lu) that are underrepresented in training, consistent with the dataset's 70 B-site elements spanning a long tail of rare compositions. The RMSE/MAE ratio of $\sim$1.6 across ensemble models indicates moderate outlier presence. The best single seed for Exp 8 achieves 0.4413 eV MAE (seed 7), while the worst achieves 0.5440 eV—a 23% spread across seeds, highlighting that even with good architecture and initialization, residual variance from dataset size remains.

The image branch's strong marginal contribution (9.7% ablation) despite using a frozen encoder pretrained on natural images suggests that even a generic visual prior extracts useful global symmetry cues. A domain-specific crystal image encoder—or equivariant 3D representations as the "visual" modality—could capture even more of this structural signal.

Overall, the experiments support three main conclusions. First, multimodal fusion improves band gap prediction consistently over unimodal baselines even in the low-data regime. Second, the specific fusion mechanism matters: lightweight conditional modulation is more effective than heavier symmetric attention architectures at this scale. Third, the trained multimodal predictor is not only accurate but operationally reusable as an oracle inside an LLM-driven inverse design loop. Together, these findings suggest that multimodal materials models can serve simultaneously as predictive scientific tools and as components within broader autonomous scientific discovery systems.

7 Conclusion and Future Directions

We presented Geo-Text-Vision (GTV), a three-modality framework for ABO_3 perovskite band gap prediction. Our best model achieves a 5-seed ensemble test MAE of 0.4862 eV—18% better than the structure-only CGCNN baseline (0.5921 eV) and 26% better than text-only SciBERT (0.6566 eV). The key architectural contribution is asymmetric FiLM modulation in which text semantically conditions the structure graph, with image features concatenated at the fusion head to bound overfitting risk. Multi-task learning with a metallicity auxiliary head provides free regularization on 383 training samples.

A systematic ablation confirms that semantic embeddings account for approximately half the total gain over the unimodal baseline, with architectural capacity explaining the other half. Contrastive ALBEF initialization reduces single-seed standard deviation (σ: 0.056 $\rightarrow$ 0.038) without improving the mean—useful variance reduction for reliable deployment. Cold start strictly outperforms warm start when the training objective changes. Multi-seed evaluation (5 seeds) is essential on a 56-sample test set, where single-seed noise ($\sigma \approx 0.05$ eV) can easily produce a false "SoTA" claim.

Beyond forward prediction, we demonstrated a zero-shot LLM-in-the-loop inverse design agent that reuses the trained ensemble as an oracle, achieving oracle errors of 0.213, 0.075, and 0.427 eV on 1.0, 2.0, and 3.0 eV targets respectively without any retraining.

Limitations. The 383-sample training set is the primary bottleneck. The text-inheritance issue in atom substitution (Robocrystallographer descriptions from the parent structure, not the swapped composition) limits fidelity of the inverse design oracle's text input. No DFT validation is performed on proposed materials.

Future work.

1. **Scale the dataset.** Relaxing space-group filters (592 $\rightarrow$ 2000+ samples) and including double perovskites ($A_2BB'O_6$) would benefit contrastive alignment, cross-attention fusion, and the inverse design loop.
2. **Domain-specific visual encoder.** Replacing frozen ImageNet-ResNet18 with an SE(3)-equivariant 3D encoder or a crystal-specific vision model would better exploit the image modality's global symmetry signal.
3. **Smarter inverse loop.** Moving beyond element substitution to crystal generative models, chemistry-tuned LLM proposers, and active learning with DFT-in-the-loop validation would convert the methodology demonstration into a materials discovery pipeline.
4. **Multi-property and multi-material prediction.** Jointly predicting formation energy, stability, and defect tolerance alongside band gap over a broader crystal family would provide richer guidance for energy-materials design.

In summary, multimodal fusion consistently outperforms single-modality baselines for perovskite property prediction. The engineering principles identified here—zero-init projections, cold start, multi-seed ensembling, ALBEF stability initialization—compose independently and are likely to generalize to other small-data multimodal regression tasks in materials science and beyond.

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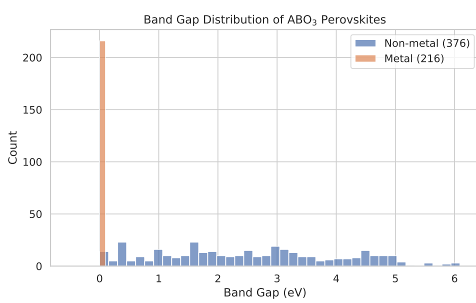
A GitHub Repositories

Each team member maintains a public GitHub repository containing all code, data splits, results, and this report:

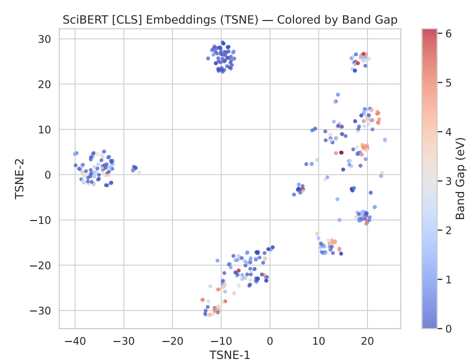
- Wenbin Ouyang: `https://github.com/oywenbin/MMA_Repo`
- Yifei Duan: `https://github.com/YifeiDuan/mmai`
- Zhuotao Jin: `https://github.com/jinzhta/MMAI_2026spring`
- Yanchen Liu: `https://github.com/liuyanchen1015/MMA`

B Additional Exploratory Data Analysis

Figure 2a shows the band gap distribution with metal/non-metal split (592 samples). The bimodal structure—a spike at 0 eV (216 metals) and a spread from 0.4 to 6 eV (376 non-metals)—motivates the metallicity auxiliary head. Figure 2b shows t-SNE of SciBERT `[CLS]` embeddings colored by band gap: even without fine-tuning, partial property-aware clustering is visible, confirming that Robocrystallographer text carries predictive signal about electronic properties.



(a) Band gap distribution



(b) SciBERT [CLS] t-SNE

Figure 2: (a) Band gap distribution with metal/non-metal split (592 samples). (b) t-SNE of SciBERT embeddings colored by band gap, showing property-aware clustering before fine-tuning.

C Full 8-Experiment Leaderboard

Table 4 gives the complete single-seed statistics for all experiments, complementing the ensemble MAE in Table 1.

Table 4: Full modality ablation leaderboard (5-seed ensemble, test set).

Exp	Method	Modality	Single-seed MAE ($\mu \pm \sigma$)	Ens. MAE	Ens. R^2
1	CGCNN	struct	0.645 ± 0.023	0.5921	0.682
2	SciBERT FT	text	0.677 ± 0.022	0.6566	0.624
3	FiLM	struct+text	0.538 ± 0.024	0.5173	0.704
5	MT-FiLM (warm)	struct+text	0.552 ± 0.016	0.5388	0.691
5b	MT-FiLM (cold)	struct+text	0.551 ± 0.056	0.4906	0.739
6	CrossAttn	struct+text	0.574 ± 0.034	0.5395	0.657
7	ALBEF init	struct+text	0.521 ± 0.038	0.4990	0.722
8	GTV (ours)	struct+text+image	0.501 ± 0.042	**0.4862**	**0.713**